Accuracy of the Scaled Flux Method for Nuclide Transmutation in Flowing-Fuel Molten Salt Reactors

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Abstract

Depletion simulations of flowing fuel molten salt reactors do not typically distinguish between in-core and ex-core regions. Instead, they either irradiate the entire primary loop while scaling the flux, or irradiate the in-core volume alone, which leaves some amount of fuel salt out of the simulation. Either approximation results in some level of inaccuracy in nuclide concentrations. The method of scaling the flux is more commonly used, but this approach assumes perfect mixing of the fuel salt in the primary loop. This work investigates a spatially resolved method to account for the impact flow and chemical removal has on concentrations of nuclides rapidly and accurately. Specifically, a 1D advection model is used which assumes no mixing in the primary loop. This method is used to provide the most conservative possible comparison, as any amount of mixing would only move the result closer to the scaled flux method. Comparisons are made in this work between this 1D advection method, the scaled flux method, other work from the literature, and experimental data. These comparisons show that the scaled flux method is sufficient for modeling most nuclides unless there is a large spatially dependent loss term, near-stationary flow rate, or artificial power oscillations with almost no mixing. In such cases, a spatially resolved method is necessary.

Keywords — Molten Salt Reactor, Molten Salt Reactor Experiment, Depletion, Chemical Removal, Flow Rate, Power Oscillations

I. INTRODUCTION

Liquid-fueled molten salt reactor (MSR) development started in the 1950s with the Aircraft Reactor Experiment [1]. Progress continued in the Molten Salt Reactor Experiment (MSRE) in the 1960s [2]. After that, solid-fueled light water reactors became the focus, which have since become the most common nuclear reactor used in the industry. However, the Generation IV International Forum in the early 2000s designated the MSR as one of the six advanced reactors of interest moving forward [3]. Since then, MSRs have garnered further interest, such as the 2020 Advanced Reactor Demonstration Program award for $113 million over seven years to the Molten Chloride Reactor Experiment.

Interest has shifted to MSRs, and many existing simulation codes and methods must adjust to account for the moving fuel. In particular, depletion simulations are of interest. In solid-fueled reactors, the fuel does not experience significant advection or diffusion of nuclides. However, MSRs have the fuel circulate through the in-core and ex-core regions. The depletion simulations must account for these regions with different power densities and the effects of circulation. Modeling this behavior accurately is vital, as the fuel composition affects reactor dynamics, reactor safety, and safeguards.

One approach is computationally inexpensive and is called the scaled flux method [4]. The scaled flux method scales the flux by the fraction of fuel salt in-core compared to ex-core. This approach does not model the in-core and ex-core regions explicitly, and instead assumes the salt is perfectly mixed in the primary loop.

Another approach has shown how spatial resolution can be investigated using a system of ordinary differential equations to solve the Bateman equation. This approach can estimate the steady-state concentration of certain nuclides across a flow loop, but it does not account for temporal resolution [5].

The toolkit MCNP/ADDER [6] has also been used to account for flow and chemical removal between neutron transport-coupled fuel depletion steps over the MSRE power history [7]. But within ADDER, ex-core loss terms (e.g., decay and chemical removal) are only treated after the calculation of the coarse depletion steps (e.g., months), as opposed to simultaneously on the timescale of the flow loop residence time (e.g., seconds). This results in an accurate removal of elements except for those nuclides that have both an ex-core loss term and a non-negligible neutron absorption cross section.

Alternatively, a more accurate treatment was recently demonstrated through MOOSE-based coupling of the deterministic neutron transport code Griffin with the thermal fluids code Pronghorn, allowing for spatial and temporal resolution of certain nuclides [8]. Tano et al. present an approach where only insoluble, medium-lived, and high concentration nuclides are tracked with pre-computed spatial distributions, which only need to be updated if the medium-lived isotopes are not in equilibrium after the thermal-hydraulics perturbation.

Spatially and temporally resolved approaches, such as the one used by Tano et al. are more computationally expensive than those that only require solving a system of ordinary differential equations. However, Tano et al. has shown that including circulation of the fuel salt in depletion directly affects the equilibrium concentration of ^{135}Xe by approximately 45% compared to a spatially unresolved approach without flux scaling [8]. This indicates there may be a need for spatially resolved depletion simulations to get accurate fuel salt compositions.

Tano et al. and Betzler have both described fully modeling advection in depletion as computationally intractable [8, 9]. This is because tracking all nuclides over space and time requires far more computational work than handling a traditional time dependent system of ordinary differential equations (ODEs). This spatially resolved system of partial differential equations (PDEs) requires accuracy in the short time scale flow effects. These small time steps cause multi-year depletion simulations to be exceedingly expensive.

However, Tano et al. showed that not all nuclides must be spatially resolved to obtain accurate results [8]. The present work builds on this idea from Tano et al. and extends a previous steady-state approach [5]. This work isolates isobars of interest and adjusts the steady-state approach by accounting for time resolution in the system of differential equations. High-fidelity thermal hydraulic simulations are computationally expensive, so instead, a 1D advection model is used to capture the in-core and ex-core effects. With the simplifications implemented, this method can run on an laptop in minutes, rather than requiring a supercomputer allocation.

Because this model does not account for any mixing, it will provide the largest difference from the scaled flux method, which uses perfect mixing. If the scaled flux and 1D advection model give the same results, then that means that both no mixing and perfect mixing will give the same results for that problem. This means that, for that category of problem, the scaled flux method can be used over a spatially resolved method with low cost and high accuracy.

II. METHODS

The ODE form of the Bateman equation written in simplified terms for nuclide i, which only contains time dependence, is given as

$$\frac{dN_i(t)}{dt} = S_i(t) - \mu_i(t)N_i(t), \tag{1}$$

$$S_i(t) = \phi(t)FY_i\Sigma_f f_{in} + \sum_{j\neq i}\lambda_j N_j(t)\gamma_{j\rightarrow i}, \tag{2}$$

$$\mu_i(t) = \lambda_i + r_i(t)f_{ex} + \phi(t)\sigma_{a,i}f_{in}, \tag{3}$$

where

$$\begin{aligned}
N_i &= \text{Concentration of nuclide } i \ [atoms\cdot cm^{-3}]\\
S_i &= \text{Source rate } [atoms\cdot cm^{-3}\cdot s^{-1}]\\
\mu_i &= \text{Loss rate } [atoms\cdot cm^{-3}\cdot s^{-1}]\\
\phi(t) &= \text{Time dependent neutron flux } [n\cdot cm^{-2}\cdot s^{-1}]\\
FY_i &= \text{Fission yield of nuclide } i \ [-]\\
\Sigma_f &= \text{Macroscopic fission cross section of the fuel } [cm^{-1}]\\
\gamma_{j\rightarrow i} &= \text{Branching ratio of } j \text{ to } i \ [-]\\
\lambda_j &= \text{Decay constant of nuclide } j \ [s^{-1}]\\
r_i &= \text{Chemical removal rate } [s^{-1}]\\
\sigma_{a,i} &= \text{Microscopic absorption cross section of nuclide } i \ [cm^2]\\
f_{in} &= \text{Fraction of fuel salt in-core } [-]\\
f_{ex} &= \text{Fraction of fuel salt ex-core } [-].
\end{aligned} \tag{4}$$

The unscaled flux method uses Equations (1), (2), and (3) where the in-core and ex-core fractions of fuel salt, f_{in} and f_{ex}, are equal to one. This is because chemical removal is applied to the entire salt volume unscaled, as is the neutron flux. The scaled flux method uses the same equation, but it applies the in-core and ex-core fractions appropriately. When one-dimensional spatial dependence is added to Equation (1) with only advection considered, a PDE is created of the form

$$\frac{\partial N_i(z,t)}{\partial t} + \nu(z,t)\frac{\partial N_i(z,t)}{\partial z} = S_i(z,t) - \mu_i(z,t)N_i(z,t), \tag{5}$$

where

$$\begin{aligned} \nu(z,t) &= \text{linear flow rate } [cm \cdot s^{-1}] \\ z &= \text{linear distance through primary loop } [cm]. \end{aligned} \tag{6}$$

Equation (5) is used in this work with a periodic boundary condition to provide the spatially resolved concentrations for target nuclides. The source term is comprised of decay sources and fission yields in this work. Sources of nuclide i from parasitic absorption are currently not considered, as the impact is assumed to be sufficiently smaller than the fission and decay parent sources. The loss term includes losses from both decay and the neutron total capture cross section in the core region. The loss term can also include a chemical removal term resolved in both space and time for nuclide i.

To solve the coupled system of equations given in Equation (5) in a computationally efficient manner, several simplifications are made. The results do not account for changes in the fuel or evolution of the neutron energy spectra [10]. Instead, these terms are assumed to be constant, which can be a reasonable approximation if fresh fuel is added and fission product poisons are removed during online reprocessing. Another simplification in this work is the use of only one spatial dimension and uniform flow. This includes simplifications to the flow of the fuel salt, which can only change in time and the single spatial dimension.

II.A. Data

Custom scripts combined with OpenMC use the linear flow rate, average neutron energy, average neutron flux, in-core fraction of fuel salt, ex-core fraction of fuel salt, in-core fuel salt volume, ex-core fuel salt volume, and average temperature to generate the appropriate cross sections and terms in Equation (5). The solver uses the average temperature and energy to generate one-group cross sections from OpenMC.

The scripts use OpenMC to access general nuclear data in ENDF-B/VII.1 [10, 11]. These data include fission yield fractions, decay chains, branching ratios, half-lives, and cross sections. These data are important for properly constructing the source and loss terms for each nuclide, while the decay chains are useful for constructing the relationships between them.

The results in this work use the data shown in Table I. The fissile concentration, neutron flux, nominal power, power history, and in-core fraction come from the MCNP/ADDER simulation of the MSRE [6, 7]. The other terms come from the data from the OpenMSR project [12].

TABLE I
Simulation parameters used in this work.

Parameter	Value	Units
Fissile Nuclide	^{235}U	-
Fissile Concentration	8.41×10^{19}	$\frac{atoms}{cm^3}$
Neutron Flux	8.4×10^{12}	$\frac{n}{cm^2 s}$
Neutron Energy	0.02533	eV
Salt Density	2.32556	$\frac{g}{cm^3}$
Temperature	294	K
Nominal Power	7.34	MW
Primary Loop Flow Length	608.06	cm
In-core Salt Fraction	27.2	%
Ex-core Salt Fraction	72.8	%
Volume	2.116111	m^3
Linear Flow Rate	21.75	$\frac{cm}{s}$

II.B. Solver

This work uses finite differencing upwind, or backwards, in space and explicit in time to generate a coupled system of PDEs. Equation (7) is used to solve for the concentration at each subsequent time step as:

$$N_k^{(l+1)} = \left(\frac{1}{1+\mu_k^{(l)}}\right)\left(N_k^{(l)} + \Delta t S_k^{(l)} + \frac{\Delta t \nu_k^{(l)}}{\Delta z}\left(N_{k-1}^{(l)} - N_k^{(l)}\right)\right), \tag{7}$$

where

$$
\begin{aligned}
N_k^{(l)} &= \text{Concentration at node } k \text{ and time } l \; [atoms \cdot cm^{-3}] \\
\mu_k^{(l)} &= \text{Loss term at node } k \text{ and time } l \; [atoms \cdot cm^{-3}] \\
S_k^{(l)} &= \text{Source term at node } k \text{ and time } l \; [atoms \cdot cm^{-3}] \\
\nu_k^{(l)} &= \text{Flow rate at node } k \text{ and time } l \; [cm \cdot s^{-1}].
\end{aligned} \tag{8}
$$

The upwind in space approach is used to maintain numerical stability [13]. This is investigated more thoroughly in Section III.C, which investigates the sensitivity of the solution to the Courant number, given in Equation (9) as:

$$
C = \frac{\Delta t \nu}{\Delta z}, \tag{9}
$$

where

$$
\begin{aligned}
\Delta t &= \text{time step } [s] \\
\Delta z &= \text{spatial grid size } [cm].
\end{aligned} \tag{10}
$$

This is important as it is used for the Courant–Friedrichs–Lewy condition, where the Courant number, C, must be less than or equal to 1 for the explicit finite differencing scheme presented in this work. Equation (9) shows that, as the spatial mesh, Δz, becomes finer or the flow rate, ν, becomes larger, the time step, Δt, must become smaller, which means higher computational cost. This is because a smaller time step becomes necessary to capture the effects spatially when the fluid moves faster through a spatial region Δz.

II.C. Evaluation Criteria

The 1D advection model provides a spatially resolved concentration profile at each point in time, while the scaled flux model provides a single concentration for each nuclide at each point in time. By comparing Equation (1) and (5), it becomes noticeable that, aside from the spatial dependence, the main difference is the existence of the advection term. As the advection term goes to zero, which could happen by the linear flow rate or the concentration gradient going to zero, then the equations closely resemble each other. In such a case, when Equation (5) is integrated over space, then the equations are equal.

Based on this observation, the metric selected to observe is the difference of each nuclide, given in Equation (11) as:

$$d_i(t) = \left| \frac{\bar{N}_{1D,i}(t) - N_{0D,i}(t)}{\bar{N}_{1D,i}(t)} \right|, \tag{11}$$

where

$$\begin{aligned} d_i(t) &= \text{Difference metric [-]} \\ \bar{N}_{1D,i}(t) &= \text{Spatially averaged 1D advection model concentration of nuclide } i \; [atoms \cdot cm^{-3}] \\ N_{0D,i}(t) &= \text{Scaled or unscaled flux model concentration of nuclide } i \; [atoms \cdot cm^{-3}]. \end{aligned} \tag{12}$$

When the difference metric approaches zero, that means that the perfect mixing model and the no mixing model give the same result. If the difference metric is non-zero, that means that the 1D advection model has a non-zero result for the advection term. For any intermediate scenario, such as inclusion of diffusion or turbulent mixing, the result can be expected to lie somewhere between the perfect mixing and no mixing results. Thus, the larger the difference metric, the greater the thermal-hydraulic modeling impact will be for that specific problem.

III. RESULTS

The results for this work include various sensitivity studies, a comparison between spatially resolved and spatially unresolved approaches, validation, and code-to-code benchmarking. Most of the results focus on ^{135}Xe, though the validation uses ^{95}Nb and ^{95}Zr. ^{135}Xe is selected for the sensitivity studies for several reasons: it has a large parasitic absorption cross section, which makes it likely to have a larger spatial dependence than other nuclides; it is important for modeling reactor behavior due to its high absorption cross section, which means evaluating it is useful; and it has a non-zero chemical removal rate, which means the chemical removal rate sensitivity can be compared with removal rates from the literature.

III.A. Spatial Sensitivity

The spatial sensitivity study informs the other analyses what level of spatial resolution is required to generate sufficiently accurate results. The time stepping sensitivity is built-in with this study, as a finer spatial mesh also requires a finer time stepping scheme due to preserving a constant Courant number of 0.5, as shown in Equation (9). This analysis compares the concentration of ^{135}Xe after one hour in the MSRE using the spatially resolved PDE method for various numbers of spatial nodes. Table II provides details on the cost and accuracy of each simulation.

TABLE II
Spatial sensitivity detailing how cost and accuracy vary based on the number of spatial nodes for ^{135}Xe with a single tracked nuclide and a Courant number of 0.5 over one hour in the MSRE.

Nodes	Cost [s]	Error [%]
50	0.28	1.81
100	0.84	1.26
200	2.76	0.82
500	16.03	0.11
1000	60.71	0.06
2000	224.43	0.01
4000	884.60	-

The computational cost scales approximately as $O(n^2)$ with the number of spatial nodes, n. Based on these results, to ensure that any small difference metric values are detectable, 500 nodes are used. This necessitates an increased computational cost, but allows for analysis of differences with uncertainty of approximately 0.01%.

III.B. Chain Sensitivity

The chain sensitivity study informs the other analyses what number of nuclides to include in the decay chain. This study compares the calculated ^{135}Xe concentration after one hour for various numbers of decay parents for ^{135}Xe included in the decay chain. Table III shows how varying the number of decay parents for ^{135}Xe in the isobar affects the results and simulation time.

The cost scales as $O(n)$ for n nuclides. Five and six nuclides have the same error because ^{135}Sn does not decay into ^{135}Sb in the OpenMC chain file used, so not including ^{135}Sn yields the same result. The expected contribution of ^{135}Sn is small regardless, so the error would also be small if it were included. Using five nuclides in the 135 isobar leading to ^{135}Xe yields accurate results, so this number of nuclides is used for the other ^{135}Xe analyses. For other analyses not

TABLE III
Chain sensitivity detailing how cost and accuracy vary based on the number of nuclides for ^{135}Xe with 500 spatial nodes and a Courant number of 0.5 over one hour in the MSRE.

Nuclides	Cost [s]	Error [%]
Xe	14.17	84.07
Xe, ^{m}Xe	28.85	58.39
Xe, ^{m}Xe, I	42.56	31.75
Xe, ^{m}Xe, I, Te	58.49	1.08
Xe, ^{m}Xe, I, Te, Sb	74.03	0.00
Xe, ^{m}Xe, I, Te, Sb, Sn	83.63	-

using ^{135}Xe, since the nuclide cost scaling is only linear, six nuclides are used for ^{95}Zr and seven for ^{95}Nb.

III.C. Courant Number Sensitivity

This sensitivity study investigates the time step size represented by the Courant, or CFL, number. A constant flow rate is used, as well as a constant number of nodes as described in Section III.A. There are two different studies used to evaluate varying the value of the Courant number. The first is a one minute MSRE simulation with 500 spatial nodes and 5 nuclides leading to ^{135}Xe. The results of this study are shown in Table IV. The results in this table show that the cost scales approximately as O(n), where n is the number of time steps. Beyond a Courant number of 1.0, the solution becomes non-physical.

TABLE IV
Courant number sensitivity study on the concentration of ^{135}Xe in the MSRE based on the time step size used with 500 spatial nodes over the course of one minute.

Courant Number	Time Steps	Cost [s]	Error [%]
1.0	2994	1.59	0.44
0.9	3327	1.77	0.48
0.8	3743	1.98	0.31
0.7	4278	2.27	0.38
0.6	4990	2.65	0.17
0.5	5988	3.19	0.28
0.4	7484	4.22	0.17
0.3	9980	5.65	0.07
0.2	14970	8.02	0.03
0.1	29940	16.03	-

The next study to investigate is a longer, one hour MSRE simulation with the other parameters unchanged from the previous study. The results of this study are shown in Table V. The

Courant number changes between these studies because the time step is calculated by dividing the increased total time by the unchanged number of time steps.

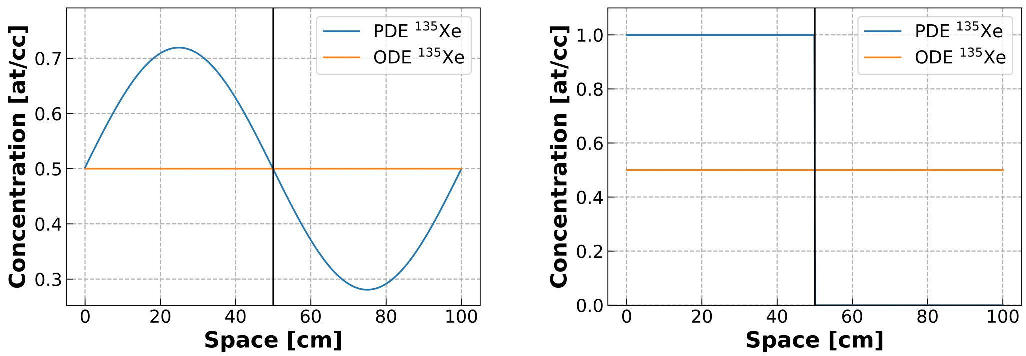
TABLE V
Courant number sensitivity study on the concentration of ^{135}Xe in the MSRE based on the time step size used with 500 spatial nodes over the course of one hour.

Courant Number	Time Steps	Cost [s]	Error [%]
1.0	179640	95.26	0.0084
0.9	199600	107.23	0.0075
0.8	224550	119.41	0.0066
0.7	256629	136.23	0.0112
0.6	299400	160.56	0.0047
0.5	359280	196.97	0.0037
0.4	449100	254.26	0.0028
0.3	598800	332.39	0.0019
0.2	898200	477.75	0.0009
0.1	1796400	956.84	-

From these studies, the error seems to consistently follow the number of time steps and depends less on the Courant number. There seems to be an inverse correlation of the net simulation time with the error for a given number of time steps, as the error for a one hour simulation was lower in comparison to all the one minute simulations with the same Courant number.

The non-monotonic variation in error with Courant number arises from the competing influences of numerical diffusion and phase accuracy introduced by changes in the time step. The phase of the low Courant number result is more accurate while the diffusion of the high Courant number result is more accurate. However, the phase of the low Courant number can be fixed by adjusting the advection rate, which is demonstrated in Figure 1.

Based on this analysis, the Courant number should be selected based on the desired behavior, particularly since the effect on the percent difference is less than half a percent for all studied cases. Because the evaluation criteria depends on the spatial gradient, preserving the sharpness of the gradient is deemed to be the most important. This means a Courant number of one is optimal for the analyses studied. Additionally, the phase shift can be corrected in a straightforward manner by adjusting the advection rate. For this work, the advection speed was found to be most accurate when multiplied by a factor of 1.002 for each case studied. There is still a slight phase shift on the first pass through the loop, but each subsequent recirculation has the expected phase.



(a) Courant number of 0.1 (b) Courant number of 1.0

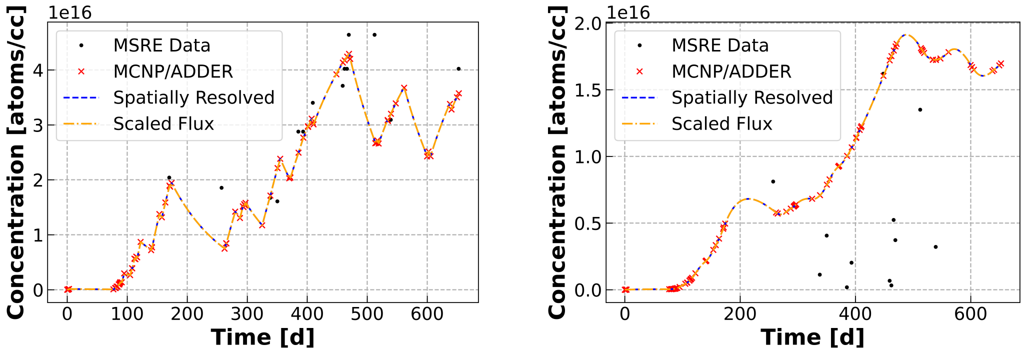
Fig. 1. Concentration for a simple test case in a 100 cm long channel with a flow rate of 10 cm/s after 5 minutes with no source or loss terms and an initial concentration in the first 50% of the loop.

III.D. Validation and Code-to-Code Benchmarking

This work uses the experimentally measured MSRE ladle salt samples over time during uranium-235 operation for validation [14]. The validation uses concentrations for ^{95}Zr and ^{95}Nb during operation in the MSRE from January 24th, 1966 to November 7th, 1967. ^{95}Zr is a salt-seeking nuclide without an appreciable noble gas precursor, which means it can be modeled without having to account for chemical removal. The ^{95}Nb acts like a noble metal in the fuel salt, so it has some chemical removal rate due to plate-out in the heat exchanger. However, fully understanding the behavior of ^{95}Nb from the existing data is challenging [15]. For this benchmark, no chemical removal will be included, though there are rates that exist in the literature for modeling removal from the fuel salt to the cover gas in the pump bowl [16].

Figure 2 shows the concentrations of ^{95}Zr and ^{95}Nb over time. In the figure, comparisons are made between the experimental results of the MSRE [14], analysis using MCNP/ADDER [7], and the current work using the spatially resolved 1D advection and scaled flux models. There appears to be reasonable agreement between the current work and the results from MCNP/ADDER. This agreement shows that the six to seven nuclides are sufficient to get a reasonable concentration for a specific target nuclide compared to a full depletion matrix.

However, the MSRE experimental data does not agree at each point, particularly for ^{95}Nb. The lack of agreement for ^{95}Zr and ^{95}Nb can be explained by uncertainty in the measurements,



(a) ^{95}Zr concentration

(b) ^{95}Nb concentration

Fig. 2. Concentrations over time in the Molten Salt Reactor Experiment with no chemical removal.

chemical effects which aren't modeled in MCNP/ADDER or this work, and other minor contributing factors these models do not include. Particularly, the ^{95}Nb concentrations from the MSRE measurements are smaller than the model predicts, which means that a non-zero chemical removal rate would likely fit these data better.

For additional code-to-code benchmarking, comparison between ^{135}Xe chain concentrations are made with a steady-state radiochemical transport model [5]. Two variations are compared, one with nine regions and one with two regions, shown in Table VI. The input data for the steady-state radiochemical model was slightly modified to match Table I. However, one difference that is likely to lead to discrepancies is the treatment of ^{135m}Xe. In this work, the source of meta-stable xenon includes an independent fission yield and from decay of iodine. In the steady-state radiochemical transport model, a cumulative fission yield was used instead without tracking the decay of iodine into metastable xenon.

TABLE VI
Spatially averaged equilibrium concentrations in the MSRE determined after three days with constant power for the time-dependent simulations via code-to-code benchmarking with steady-state model [5].

Method	Nuclide	N_i $[atoms \cdot cm^{-3}]$	d_i [%]
1D Advection	^{135}Xe	2.69×10^{14}	-
Scaled Flux	^{135}Xe	2.69×10^{14}	0.0
Two Region [5]	^{135}Xe	3.16×10^{14}	17.3
Nine Region [5]	^{135}Xe	2.58×10^{14}	4.1
1D Advection	^{135m}Xe	1.81×10^{12}	-
Scaled Flux	^{135m}Xe	1.81×10^{12}	0.0
Two Region [5]	^{135m}Xe	1.55×10^{12}	14.1
Nine Region [5]	^{135m}Xe	1.57×10^{12}	13.4
1D Advection	^{135}I	2.41×10^{14}	-
Scaled Flux	^{135}I	2.41×10^{14}	0.0
Two Region [5]	^{135}I	2.65×10^{14}	9.7
Nine Region [5]	^{135}I	2.35×10^{14}	2.5
1D Advection	^{135}Te	1.04×10^{11}	-
Scaled Flux	^{135}Te	1.04×10^{11}	0.0
Two Region [5]	^{135}Te	9.22×10^{10}	11.0
Nine Region [5]	^{135}Te	9.59×10^{10}	7.5
1D Advection	^{135}Sb	3.97×10^{8}	-
Scaled Flux	^{135}Sb	3.97×10^{8}	0.0
Two Region [5]	^{135}Sb	3.69×10^{8}	7.1
Nine Region [5]	^{135}Sb	3.83×10^{8}	3.7

Table VI shows that there is good agreement between the 1D advection model and scaled flux model for all nuclides tracked, with the largest difference being approximately 10^{-3}% for ^{135}Xe. When comparing with the two and nine region models, in every case the nine region model has a smaller difference. This is likely because the level of spatial refinement is finer in the nine region case, which brings it closer to the more spatially refined model presented in this work. In the nine region model, ^{135}Sb is 3.7% smaller, ^{135}Te is 7.5% smaller, ^{135}I is 2.5% smaller, ^{135m}Xe is 13.4% smaller, and ^{135}Xe is 4.1% smaller. In the two region model, ^{135}Sb is 7.1% smaller, ^{135}Te is 11% smaller, ^{135}I is 9.7% larger, ^{135m}Xe is 14.1% smaller, and ^{135}Xe is 17.3% larger. Overall, there is reasonable agreement between the steady-state models and the current time-dependent models. The larger differences in ^{135m}Xe are expected, and the other nine region results are within 10%.

III.E. Difference Sensitivities

Based on Equations (1) and (5), the difference metric described in Section II.C can be manipulated by altering the source term, S_i; loss term, μ_i; and flow rate, ν. If the manipulations designed to lead to large differences result in a negligible difference, then that would prove that those scenarios, or any more conservative scenarios, do not require spatial resolution. In this work, artificial manipulation of nuclear data, such as fission yields or branching ratios, will not be considered. Instead, manipulations will be performed on properties which can be more readily controlled: the neutron flux, the chemical removal rate, and the fuel salt flow rate.

III.E.1. Flux Sensitivity

To create a gradient in the concentration through flux manipulation, the most direct way is through power oscillations in time. Specifically, a square wave can be used such that a certain portion of the fuel salt is irradiated at a high flux while another portion of fuel salt receives no irradiation. This represents the most extreme case of a spatial gradient which can be formed using flux variations. Since a realistic model would have some amount of mixing and a lesser amount of power oscillation, this provides a limiting case. If the difference metric is close to zero, then that would indicate that for power oscillation scenarios, spatial resolution is generally not necessary. A maximum difference less than 1% over tens of circulations around the primary loop would indicate that, with mixing, any realistic molten salt reactor could have power oscillations modeled during a depletion simulation without spatial resolution.

To simplify the problem, the data is adjusted to a flow rate of 10 cm/s over 100 cm, with 50% of the salt in-core and 50% ex-core. Figure 3 shows how the concentration evolves over time. This figure normalizes the concentration at each time step by the peak concentration at that time step. This leads to the diagonal striping effect, as the peak concentration advects throughout the primary loop.

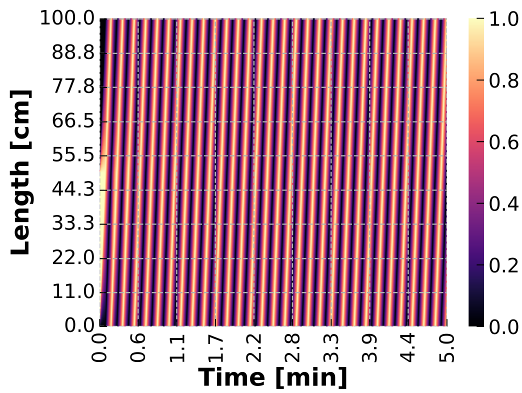
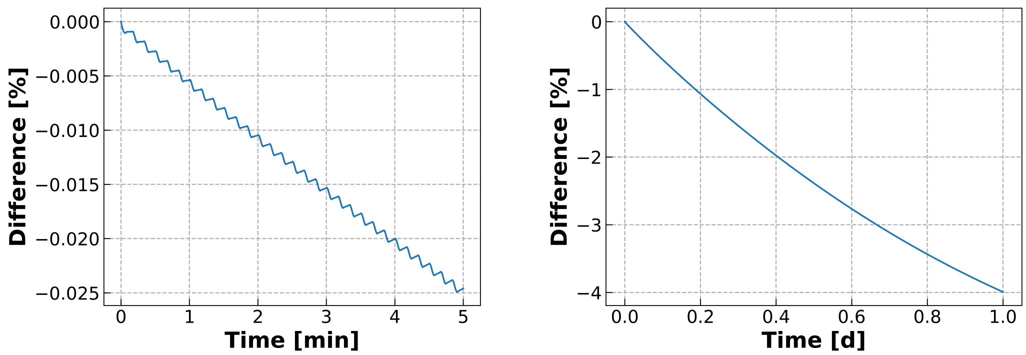


Fig. 3. Relative maximum concentration at each time step of ^{135}Xe in a toy model with 5 second in-core and ex-core residence times and a square wave power oscillation.

Figure 4 shows how the difference metric evolves over a short period of time and over a longer period of time for ^{135}Xe. The five minute long simulation is sufficiently short such that the step-wise shape of the power oscillations become visible in the difference evolution via a shape with distinct steps down after each recirculation. As the fuel circulates over the first five minutes, the staircase shape trends linearly downwards, though the flatness of each portion begins to become an upwards trend. The reason for this is because of the diffusion-inducing behavior of the precursors to ^{135}Xe. Because these other nuclides decay into ^{135}Xe at different locations, the large gradient caused by the power oscillations becomes lessened. When modeled with only a single nuclide, the general trend is the same but without positive slopes.

These results show that it takes approximately 5 hours, or approximately 15,000 circulations around the primary loop, before the difference metric reaches 1%. This means that, provided there is some small amount of mixing via turbulence or diffusion, the scaled flux method is sufficiently accurate to model the ^{135}Xe in the MSRE. In the case that there is no mixing, or the level of mixing is negligible, then a spatially resolved approach may be necessary if there are power oscillations which occur at such a frequency that leads to heterogeneity in the fuel salt.



(a) 5 minutes - 30 recirculations

(b) 1 day - 8640 recirculations

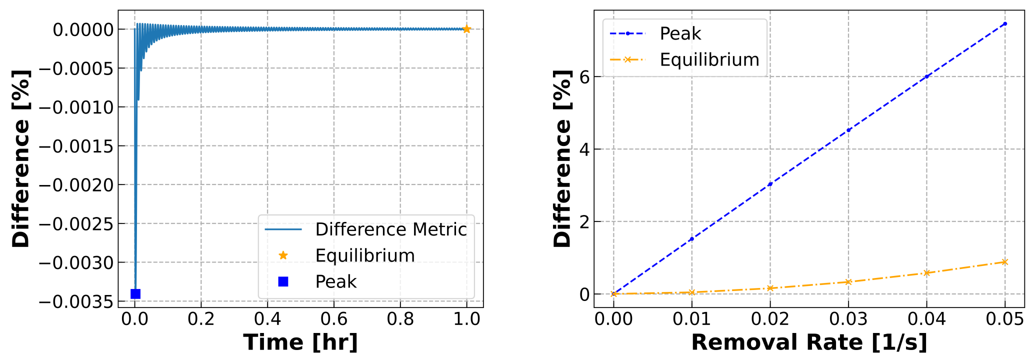
Fig. 4. ^{135}Xe difference metrics in a toy model with 5 second in-core and ex-core residence times over different periods of time.

III.E.2. Chemical Removal Sensitivity

In this work, the chemical removal is applied on each spatial node that exists in the ex-core region. The xenon chemical removal rate from the pump bowl via the off-gas system in the MSRE is $4.067 \times 10^{-5}\ s^{-1}$ [17]. For the Molten Salt Breeder Reactor (MSBR), the chemical removal rate of xenon is $0.05\ s^{-1}$ [18]. The scenario in which the largest spatial gradient would exist is one with a very large chemical removal rate in the ex-core region. A large chemical removal rate leads to the largest spatial gradient because as the chemical removal rate goes to infinity, the concentration would rapidly approach 0 in that region.

Figure 5 shows how the difference metric of ^{135}Xe evolves over time and with varying removal rate. The difference rate has a sinusoidal shape that exponentially decays, with a peak difference early on and than reaching an equilibrium difference. The peak and equilibrium differences are plotted for varying removal rate values of xenon.

Based on the xenon removal rate in the MSRE, the difference metric is negligible, meaning the scaled flux model can be used with high accuracy. For the xenon removal rate in the MSBR, the difference peaks at 7.5% and is at equilibrium at 0.9%. This means that depletion simulations of the MSBR may need spatial resolution during depletion simulations to accurately model ^{135}Xe concentrations.



(a) Difference metric with no chemical removal with peak and equilibrium marked

(b) Peak and equilibrium difference metrics for various chemical removal rates

Fig. 5. ^{135}Xe chemical removal rate difference metrics in the MSRE.

III.E.3. Flow Rate Sensitivity

The flow rate is inversely proportional to the residence time of the salt in a given region. A faster flow rate, or shorter residence time in each region, is expected to lead to a smaller difference. This is because the salt would be more evenly irradiated, leading to a more homogeneous salt. As the residence time increases, the spatial gradient is expected to increase.

However, as can be seen in Equation (5), the gradient is multiplied by the flow rate. This means that there can be expected to be competing behavior from these terms. For a stationary fuel salt, the scaled flux model would no longer be correctly handling the physics, as the system is stationary and would be better modeled by the unscaled flux method. This means that, even with competing behavior, the trend should be towards an increase in the difference metric with decreasing flow rate.

Figure 6 shows how the difference metric for ^{135}Xe changes for various residence times, calculated by dividing the length of the channel by the flow rate. The peak and equilibrium difference metrics both trend upwards as the residence time increases, which aligns with expectations. This means that for typical MSRE depletion simulations with a residence time of approximately 30 seconds, the scaled flux method has a difference metric less than 0.01% due to fuel salt advection. However, if a pump coast-down or some other loss of flow condition is part of the simulation, then it will require spatial resolution to properly model the ^{135}Xe.

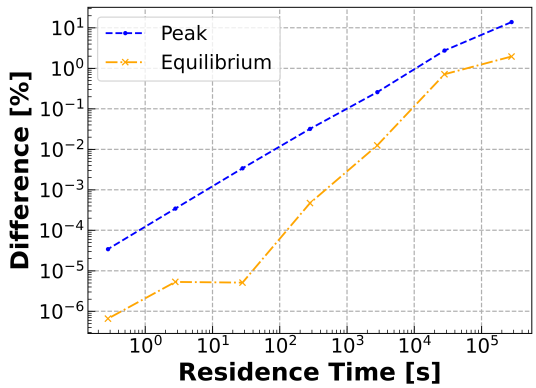


Fig. 6. Peak and equilibrium difference metrics for various primary loop residence times in the MSRE for ^{135}Xe.

IV. CONCLUSIONS

This work compares a scaled flux model, which assumes perfect mixing, and a spatially resolved 1D advection model, which assumes no mixing, to determine the maximum error that could be expected from using the scaled flux method. ^{135}Xe was selected to be the main target of investigation as it has a large absorption cross section, and thus would be more affected by the spatial resolution than other nuclides. The advection and scaled flux models offer close agreement with MCNP/ADDER, while the data from the MSRE for ^{95}Zr is in reasonable agreement.

The analyses in this work demonstrated the sensitivity of the difference between the scaled flux and advection models to flux, chemical removal rates, and flow rates. For the flux, in general, minor variations in the flux can be modeled using the scaled flux method. If the power is artificially oscillated to induce heterogeneity and the MSR has little to no mixing, then the scaled flux method would no longer be sufficient. For chemical removal rates, small removal rates can be handled using the scaled flux method while large chemical removal rates, such as xenon in the MSBR, may require spatial resolution. For flow rates, the scaled flux method becomes increasingly inaccurate as the flow rate decreases.

Future work could involve continued verification and validation efforts for isotopes that exhibit in-core and ex-core loss terms due to neutron capture and chemical removal, respectively. Future work could also compare these methods to a modified-point-kinetics-like approach that treats the ex-core region with an exponential decay, offering an approach between the 0-D and 1-D approaches as shown in MacPhee [19]. Finally, future work of interest could include investigation of depletion simulations of various MSRs in the literature and analysis on nuclide concentration difference metrics.

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